

Falsifiability rehearsal — §6.N.1.3 companion to real-device- verification.{md,json}

Per the pre-push hook's §6.N.1.3 gate, this file records a genuine, executed falsifiability rehearsal of the production code path that `real-device-verification.{md,json}` (commit `c1b13ca7`) claims to cover — specifically Task 3 (`Challenge00CrashSurvivalTest`), the load-bearing cold-start-survival canary that PASSEd in that attestation.

Target

`Challenge00CrashSurvivalTest.continueOnArchive0rg_persists_signal_eAuthState_to_disk(app/src/androidTest/kotlin/lava/app/challenges/Challenge00CrashSurvivalTest.kt)`, whose own KDoc already documents this exact rehearsal (lines 50-65). This record is the **genuine execution** of that documented rehearsal against the same containerized-emulator matrix path (`scripts/run-challenge-matrix.sh, --runner=containerized, --avds "CZ_API34_Phone:34:phone"`) that produced the attestation being gated.

Mutation

`core/auth/impl/src/main/kotlin/lava/auth/impl/AuthServiceImpl.kt`, `signalAuthorized()` — commented out:

```
preferencesStorage.saveSignaledAuthState(name = name, avatarUrl =  
    avatarUrl)
```

Step 1 — Rebuild + run against MUTATED code: FAIL (as expected)

```
$ export JAVA_HOME=/home/milosvasic/t062_work/jdk17/  
jdk-17.0.20.1+1  
$ ./gradlew :app:assembleDebug :app:assembleDebugAndroidTest --no-
```

```

daemon
BUILD SUCCESSFUL in 52s

$ ./scripts/run-challenge-matrix.sh --no-build \
  --avds "CZ_API34_Phone:34:phone" \
  --test-class
"lava.app.challenges.Challenge00CrashSurvivalTest" \
  --evidence-dir ".lava-ci-evidence/2026-09-22-rehearsal-
MUTATED-signalAuthorized" \
  --boot-timeout 8m --test-timeout 8m
...
[1] CZ_API34_Phone          api=34 phone FAIL
      error: exit status 1
MATRIX FAILED – at least one AVD did not pass. tag.sh MUST refuse
this commit.

```

Observed failure (CZ_API34_Phone/gradle.log:1242-1245, full log preserved at .lava-ci-evidence/2026-09-22-rehearsal-MUTATED-signalAuthorized/):

```

lava.app.challenges.Challenge00CrashSurvivalTest >
continueOnArchiveOrg_persists_signaledAuthState_to_disk[sdk_gphone6
- 14] FAILED
      java.lang.AssertionError: PRIMARY: SignaledAuthState MUST be
persisted on disk after Continue tap. Phase 1.1 production-stack
guarantee for force-stop survival.

```

Tests on sdk_gphone64_x86_64 - 14 failed: There was 1 failure(s).

Machine attestation: .lava-ci-evidence/2026-09-22-rehearsal-MUTATED-signalAuthorized/real-device-verification.json —
 "all_passed": false, "test_passed": false.

(Note: a first attempt at this step failed with an unrelated 25.0.4 Gradle/JDK error because JAVA_HOME was not exported for that invocation — a host-toolchain issue, not the rehearsal's mutation. Re-run with JAVA_HOME correctly exported produced the genuine assertion failure above. Recorded honestly per §11.4.6.)

Step 2 — Revert

```

$ git diff --stat core/auth/impl/src/main/kotlin/lava/auth/impl/
AuthServiceImpl.kt
(empty – byte-identical to pre-mutation)

```

Step 3 — Rebuild + run against REVERTED code: PASS

```
$ ./gradlew :app:assembleDebug :app:assembleDebugAndroidTest --no-daemon
BUILD SUCCESSFUL in 50s
```

```
$ ./scripts/run-challenge-matrix.sh --no-build \
  --avds "CZ_API34_Phone:34:phone" \
  --test-class
"lava.app.challenges.Challenge00CrashSurvivalTest" \
  --evidence-dir ".lava-ci-evidence/2026-09-22-rehearsal-
REVERTED-signalAuthorized" \
  --boot-timeout 8m --test-timeout 8m
...
[1] CZ_API34_Phone          api=34 phone PASS
MATRIX PASSED — every AVD booted and every test passed.
```

Machine attestation: .lava-ci-evidence/2026-09-22-rehearsal-REVERTED-signalAuthorized/real-device-verification.json —
"all_passed": true, "test_passed": true.

Verdict

GENUINE. Challenge00CrashSurvivalTest fails with a clear, specific assertion message when the exact production code path it claims to cover (AuthServiceImpl.signalAuthorized persisting SignaledAuthState to disk) is broken, and passes again once reverted. The attestation in real-device-verification.{md,json} (commit c1b13ca7) is not a bluff.

Step	Result	Evidence dir
Mutated	FAIL — AssertionError:	.lava-ci-evidence/
	PRIMARY: SignaledAuthState MUST be persisted...	2026-09-22-rehearsal- MUTATED-signalAuthorized/
Reverted	PASS	.lava-ci-evidence/ 2026-09-22-rehearsal- REVERTED- signalAuthorized/